

Want to find r such that $f(r) = 0$

(Using Dual numbers)

$$f(r) = 4\pi r - \frac{2}{r^2}$$

```
f(r)=4π*r-2.0*(1/r)*(1/r)
r=Dual(1.0,1.0)

for i=1:50
    r=r-f(r).v/f(r).d
end
```

$$\left(r = 1, \frac{\partial r}{\partial r} = 1 \right)$$

$$r_{i+1} = r_i - f(r_i)/dfdr(r_i)$$

Want to find r such that $f(r) = 0$
(Using Dual numbers)

Want to find r such that $f(r) = 0$
(Using ForwardDiff package)

```
f(r)=4π*r-2.0*(1/r)*(1/r)
r=Dual(1.0,1.0)

for i=1:50
    r=r-f(r).v/f(r).d
end
```

$$f(r) = 4\pi r - \frac{2}{r^2}$$

$$\left(r = 1, \frac{\partial r}{\partial r} = 1 \right)$$

$$r_{i+1} = r_i - f(r_i)/dfdr(r_i)$$

```
f2(x)=4π*x-2/x^2
x=1.0
for i=1:10
    x-=f2(x)/ForwardDiff.derivative(f2,x)
    println(x)
end
```


$$f_1(\vec{x}) = x_1^2 - x_2 + 1 = 0$$

$$f_2(\vec{x}) = 3 \cos(x_1) - x_2 = 0$$

$$\begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix} \begin{Bmatrix} x_1^{(i+1)} - x_1^{(i)} \\ x_2^{(i+1)} - x_2^{(i)} \end{Bmatrix} = - \begin{Bmatrix} f_1(x_1^{(i)}, x_2^{(i)}) \\ f_2(x_1^{(i)}, x_2^{(i)}) \end{Bmatrix}$$

```
xvals=[1,1]
for i=1:10
    F=VectorFunc(xvals)
    J=hcat(getfield.(VectorFunc([Dual(xvals[1],1.0),Dual(xvals[2],0.0)]))[:],:∂),
          getfield.(VectorFunc([Dual(xvals[1],0.0),Dual(xvals[1],1.0)]))[:],:∂))
    delta=-J\F
    xvals+=delta
    println(xvals)
end
```

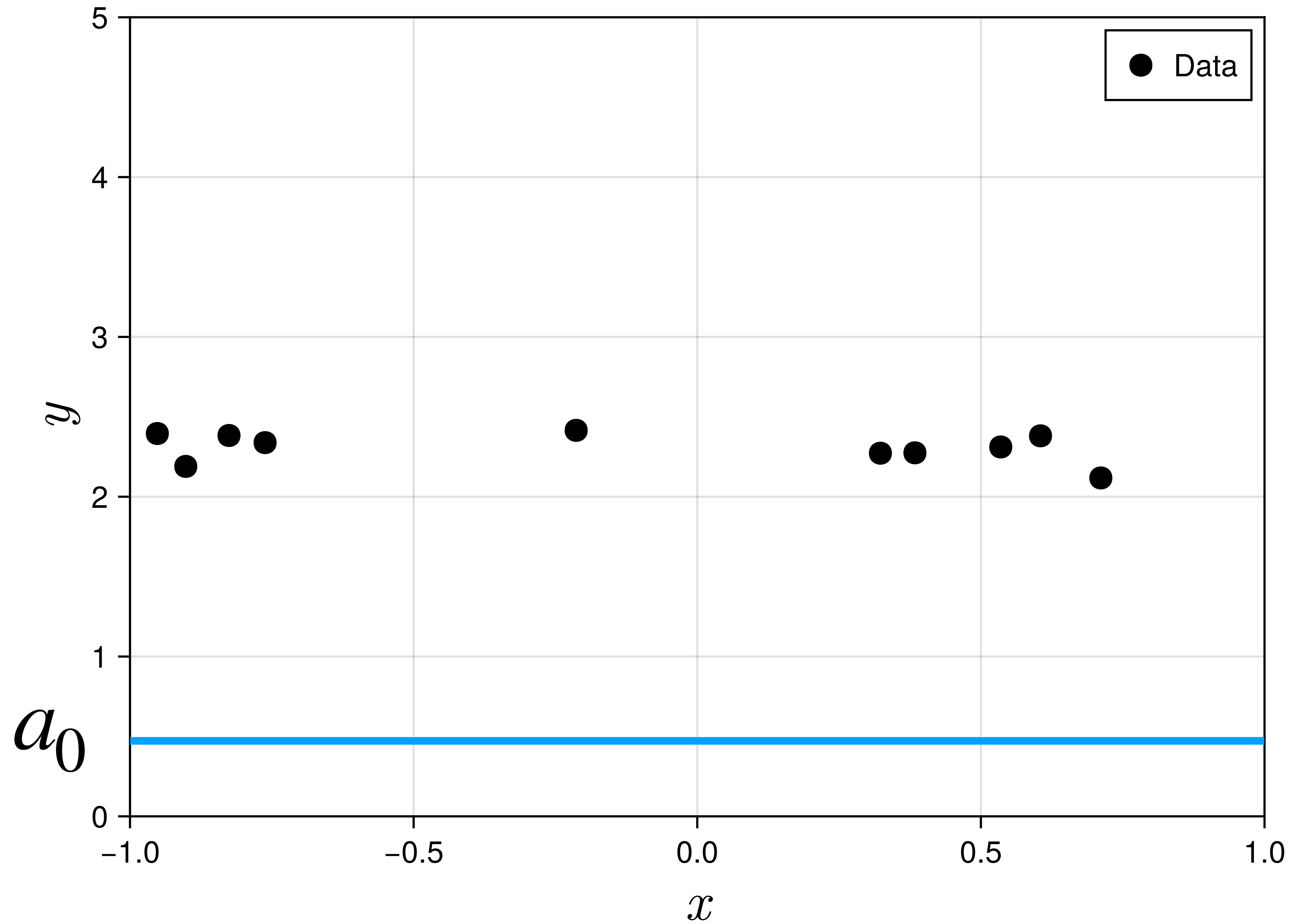
$$\text{VectorFunc}(x) = [x[1]^2 - x[2] + 1, \\ 3 * \cos(x[1]) - x[2]]$$

Code using your own Forward Mode Automatic Differentiation

```
xvals=[1,1]
for i=1:10
    F=VectorFunc(xvals)
    J=hcat(getfield.(VectorFunc([Dual(xvals[1],1.0),Dual(xvals[2],0.0)])[:],:∂),
getfield.(VectorFunc([Dual(xvals[1],0.0),Dual(xvals[1],1.0)])[:],:∂))
    delta=-J\F
    xvals+=delta
    println(xvals)
end
```

Code using Julia's ForwardDiff() package

```
x=[1,1]
for i=1:5
    F=VectorFunc(x)
    J=ForwardDiff.jacobian(VectorFunc,x)
    delta=-J\F
    x+=delta
    println(x)
end
```

$$f_{model} = a_0$$

$$S(a_0) = \frac{1}{2} \sum_{i=1}^N (y_i - a_0)^2$$

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```
S(a0)=0.5*sum((yi.-a0).^2)
```

We can find Gradient of S with Dual numbers

at $a_0 = 1.0$

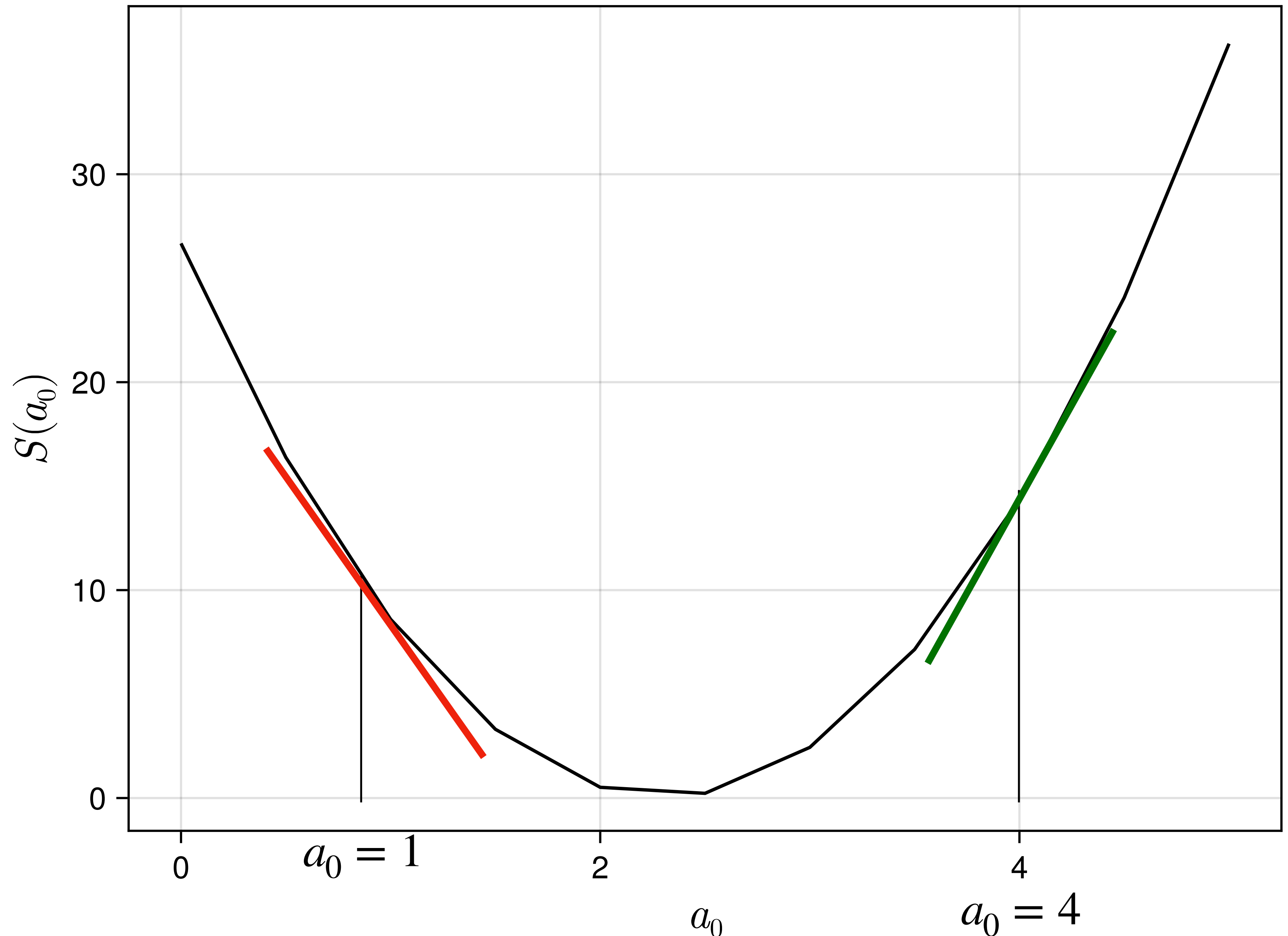
```
a0=Dual(1.0,1.0)
S(a0).d
```

-13.079059040647497

at $a_0 = 4.0$

```
a0=Dual(4.0,1.0)
S(a0).d
```

16.920940959352503



Standard Gradient Descent

```
a0=Dual(1.0,1.0)
α=0.01

for i=1:500
    a0.v=a0.v-α*S(a0).∂
    println(a0.v)
end
```

$$\frac{\partial S}{\partial a_0}$$

$$S(a_0) \cdot \partial$$